Assignment 2

Question 1:

Scenario: A researcher uses linear regression to predict whether a tumor is malignant (1) or benign (0). The predictions are all fractional values between 0.2 and 0.8.
Questions:
1. Why is linear regression not suitable for this problem?
2. What does logistic regression do differently?
3. What does the sigmoid function represent in this context?

Question 2:

Scenario: A company uses kNN for classifying customer segments. When new data is added, predictions suddenly change drastically.
Questions:
1. Why might kNN behave this way?
2. What hyperparameter could stabilize its predictions?
3. How does distance metric choice affect results?

Question 3:

Scenario: A classifier gives high accuracy but fails to detect rare fraud cases.
Questions:
1. Why might accuracy be misleading here?
2. Which other metric should be used?
3. How do ROC and AUC help understand model performance?